

Einstein’s Heuristic as Statistical Inference: The Photoelectric Effect as Machine Learning, 1887–1916

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Abstract

In his 1905 “miraculous year” paper on the photoelectric effect, Albert Einstein proposed that electromagnetic radiation consists of discrete energy quanta $h\nu$. His argument had two components: a physical *model* (the light-quantum hypothesis) and an implied *measurement equation* relating the observable stopping potential V_0 linearly to the incident frequency ν . This note examines Einstein’s reasoning from a modern statistical perspective. I assemble the historical data available in 1902–1916—Lenard’s qualitative findings, the intermediate quantitative work of Hughes (1912) and Richardson & Compton (1912), and Millikan’s definitive measurements (1916)—and carry out the simple ordinary-least-squares regression that the theory implies: $V_0 = \alpha + \beta\nu + \varepsilon$. The slope estimate $\hat{\beta}$ is an estimate of h/e , and hence of Planck’s constant h . Millikan’s five-point regression gives $\hat{\beta} = 4.129 \times 10^{-15} \text{ V}\cdot\text{s}$, implying $h = 6.61 \times 10^{-34} \text{ J}\cdot\text{s}$ (0.5% above the modern value). A maximum likelihood treatment under Gaussian errors recovers the same point estimates and establishes that OLS achieves the Cramér-Rao bound; the bound also provides a formal justification for Millikan’s design choice to use a wide frequency range. Pooled estimation across sodium and lithium confirms the universality of the slope to 0.2%. I interpret the episode through the lens of modern machine learning: Einstein’s quantum hypothesis supplied the hypothesis class and inductive bias; Lenard’s feature-importance finding eliminated intensity as a relevant input; and the final regression is a textbook supervised-learning problem in which five training points suffice because the physics has already fixed

the functional form. The contrast with a purely data-driven learner illuminates both the promises and the limits of theory-free AI for scientific discovery.

1 Introduction

In December 1900, Max Planck introduced the energy quantum $h\nu$ as “an act of desperation” to fit the blackbody spectrum. Five years later, Einstein borrowed that quantum and applied it to a second phenomenon—the photoelectric effect—in a paper that would earn him the 1921 Nobel Prize. Einstein’s paper posited a falsifiable linear equation relating the stopping potential of emitted electrons to the incident frequency, with slope h/e —a prediction that Millikan confirmed to 0.5% in 1916. It also introduced *probability* into physics: if light consists of discrete quanta whose average spatial density tracks the classical wave intensity $|E|^2$, then $|E|^2$ functions as a probability density—not merely a classical energy density. Max Born made this logic the cornerstone of quantum mechanics in 1926 [Born, 1926]: his rule $|\psi|^2 = \text{probability density}$, now called Born’s rule, was—as Born stated explicitly—an extension of Einstein’s photon reasoning to the general Schrödinger wave function. The paper that launched the precision measurement effort documented here thus led to Born’s rule. Einstein spent the rest of his life opposing it.

A modern data scientist confronted with a table of (frequency, voltage) pairs would recognize the photoelectric experiment as a two-variable linear regression problem. The regression came after the theoretical insight. Albert Einstein in 1905 wrote down the model

$$V_0 = \frac{h}{e} \nu - \frac{W}{e} \tag{1}$$

before anyone had measured a reliable V_0 - ν slope. Equation (1) is a prediction from what Planck himself called “simply an act of desperation”¹: that light arrives in discrete quanta of energy $h\nu$. The slope coefficient is h/e , where h is Planck’s constant and e the elementary charge. The intercept is $-W/e$, where W is the metal’s work function—the energy cost of extracting an electron from the bulk.

¹Planck introduced the energy quantum $h\nu$ in December 1900 to derive the blackbody radiation spectrum, later describing his own step in a letter to R. W. Wood (7 October 1931): “Briefly summarized, what I did can be described as simply an act of desperation... a theoretical interpretation had to be found at any cost, no matter how high.” [Planck, 1931]. Einstein borrowed the quantum $h\nu$ five years later and applied it to photoemission.

Eleven years later, Robert Millikan [Millikan, 1916] provided the definitive empirical confirmation, using freshly scraped alkali-metal surfaces in high vacuum and narrow-band filtered spectral lines from a mercury arc. His data, assembled in Table 3 below, allow a textbook OLS regression that recovers h to within 1% of its modern value.

The episode provides a microcosm of scientific inference:

1. **Model specification.** Einstein’s quantum hypothesis imposes linearity of V_0 in ν , a slope h/e that is the same for every metal, and a metal-specific intercept.
2. **Experimental design as data quality control.** Contaminated surfaces, stray light, and a restricted frequency range biased the estimates of Lenard, Hughes, and Richardson & Compton. Millikan solved these problems one by one.
3. **Parameter identification.** Variation in ν across widely separated spectral lines identifies the slope h/e . The intercept identifies the work function W only up to the contact electromotive force (EMF) between the photoemitter and the collector, a nuisance parameter that shifts the intercept but not the slope.
4. **Reluctance to accept the model.** Millikan confirmed Einstein’s equation while publicly rejecting the light-quantum interpretation.
5. **Unintended conceptual consequences.** The light-quantum hypothesis makes the classical wave intensity $|E|^2$, reinterpreted as a photon density, a probability density. Born traced his probabilistic interpretation of the Schrödinger wave function to this precedent [Born, 1926].

Einstein’s quantum hypothesis committed him to a view in which $|E|^2$ governs the statistical behavior of photons rather than the energy of a continuous wave. In private correspondence Einstein called this a *Gespensterfeld* (ghost field): a wave that guides probabilities without itself being physically real in the classical sense. Born’s 1926 paper [Born, 1926] formalised that idea for the Schrödinger wave function. Einstein won the 1921 Nobel Prize for the equation. Born won the 1954 Nobel Prize for the probabilistic interpretation that the equation implied.

Section 2 sketches the experimental history. Section 3 develops the regression framework. Section 4 assembles the historical data. Section 5 presents the OLS estimates and diagnostics. Section 6 develops maximum likelihood estimation of Einstein’s parameters and derives the Cramér-Rao bound. Section 7 discusses the contact-EMF identification

problem. Section 8 analyses Millikan’s nine two-point determinations. Section 9 compares historical estimates of h . Section 10 distinguishes inference from interpretation. Section 11 reinterprets the episode in the language of machine learning and AI. Section 12 contrasts Einstein’s regression with what a deep neural network would learn from the data available at each epoch. Section 13 concludes.

2 Experimental History, 1887–1916

2.1 Discovery and Qualitative Characterisation

Heinrich Hertz observed in 1887 that ultraviolet light reduced the sparking distance in a gap discharge. Wilhelm Hallwachs (1888) showed that a zinc plate illuminated by UV light lost negative charge. These observations established that light could eject negatively charged particles from metals.

J. J. Thomson (1899) identified the ejected particles as electrons by measuring their charge-to-mass ratio. The same year Philipp Lenard began systematic experiments, culminating in his landmark 1902 paper [Lenard, 1902].

2.2 Lenard’s Key Finding (1902)

Lenard’s decisive contribution was establishing two qualitative facts:

1. The *maximum kinetic energy* (equivalently the stopping potential V_0) of the emitted photoelectrons is **independent of light intensity**. Classical wave theory predicted the opposite: more intense waves deliver more energy.
2. V_0 depends on the **frequency** (“colour”) of the incident light, with higher frequency producing higher V_0 .

Lenard’s quantitative data were, by his own account, rough: his metal surfaces were oxidised, vacuum conditions poor, and no attempt was made to separate the dependence of V_0 on ν from confounding effects. He received the 1905 Nobel Prize in Physics for his cathode-ray work.

2.3 Einstein’s 1905 Hypothesis

In March 1905 [Einstein, 1905] Einstein proposed that each light quantum carries energy $h\nu$. A single quantum is absorbed by a single electron. The electron escapes if $h\nu > W$; the excess appears as kinetic energy, so the stopping potential satisfies equation (1). At the time of writing, Einstein noted:

“As far as I can see, our conception is not at variance with the properties of the photoelectric effect observed by Mr. Lenard.”

There were, however, no quantitative measurements of the V_0 – ν slope. The predicted slope h/e is a precise, falsifiable number: 4.136×10^{-15} V·s (using modern constants).

2.4 Early Experiments (1907–1914)

Between 1907 and 1914 three groups attempted quantitative verification. Table 6 summarises their estimates of h .

Ladenburg (1907). First quantitative V_0 – ν measurements, but over a wavelength range of only ~ 50 nm (2000–2600 Å). Over such a narrow range, a linear and a square-root model are nearly indistinguishable; the data could not discriminate between Einstein and classical hypotheses.

Hughes (1912). Arthur Llewelyn Hughes [Hughes, 1913] used three UV lines ($\lambda = 1849, 2257, 2537$ Å) and roughly a dozen metals. Two of the three frequencies were used to fit the line; the third served as a check. Hughes reported h values ranging from 3.55×10^{-27} to 5.85×10^{-27} erg·s—a spread of 65%. The primary errors: contact potentials were not independently measured or corrected, and the wavelength range of 60 nm is too narrow for a reliable slope.

Richardson & Compton (1912). Owen Richardson and Karl Compton [Richardson and Compton, 1912] measured eight metals. Their h estimates varied by more than 60%. Millikan [1916] attributed the scatter to the same deficiencies as Hughes: narrow frequency range, uncontrolled contact EMF, and stray short-wavelength scattered light.

2.5 Millikan’s Definitive Measurement (1916)

Robert Millikan spent a decade developing his “machine shop in vacuo.” Inside an evacuated glass bulb, an electromagnetically driven knife freshly shaved the surface of a cast cylinder of sodium, potassium, or lithium. The wheel rotated the fresh surface into the light beam without breaking vacuum. The contact EMF was independently measured by the Kelvin double-balance method and monitored throughout each run. Only spectral lines with no short-wavelength companions in the mercury arc were used, and those lines were further filtered to remove scattered UV.

The result was the regression in Section 5.

3 The Regression Equation

3.1 Derivation

Einstein’s light-quantum hypothesis and the energy-conservation condition for photo-electron escape give

$$\underbrace{eV_0}_{\text{max KE}} = \underbrace{h\nu}_{\text{photon energy}} - \underbrace{W}_{\text{work function}}, \quad (2)$$

where all quantities are in SI units. Dividing by e gives the observation equation

$$V_0 = \underbrace{\frac{h}{e}}_{\beta} \nu - \underbrace{\frac{W}{e}}_{\alpha_0} + \underbrace{v_c}_{\text{contact EMF}} + \underbrace{\varepsilon}_{\text{meas. error}}, \quad (3)$$

where v_c is the contact EMF (a nuisance parameter that shifts the intercept but not the slope) and ε is measurement noise. Let $\alpha \equiv -W/e + v_c$. The estimating equation is

$$\boxed{V_0 = \alpha + \beta \nu + \varepsilon, \quad \beta = \frac{h}{e}.} \quad (4)$$

This is a simple bivariate linear regression with one unknown slope (β) and one unknown intercept (α).

3.2 Identification

The slope. Variation in ν across spectral lines identifies $\beta = h/e$. The standard error of $\hat{\beta}$ is $s/\sqrt{S_{\nu\nu}}$, where $S_{\nu\nu} = \sum_i (\nu_i - \bar{\nu})^2$; spreading the ν_i apart raises $S_{\nu\nu}$ and shrinks the standard error. The early experiments were confined to a 60 nm UV range and gave unreliable estimates.

The intercept and the contact-EMF problem. The intercept α conflates the work function W and the contact EMF v_c . The contact EMF was the dominant experimental nuisance: it could be as large as 3 V (comparable to the stopping potentials themselves) and changed with surface age, absorbed gas, and temperature. The term v_c enters additively and shifts all stopping potentials by the same amount. A v_c that stays constant during a run leaves β unaffected. Millikan verified this by monitoring v_c continuously; he found it stable to ± 0.01 V over several hours.

The universal slope as a cross-metal restriction. Because $\beta = h/e$ is a universal physical constant, equation (4) implies that *all* metals share the same slope. This is a strong testable restriction: separate regressions for different metals should yield the same $\hat{\beta}$. Millikan confirmed it: his sodium and lithium slopes agreed to within 0.2%.

4 The Historical Data

4.1 Millikan's Mercury Arc Lines

Millikan used isolated lines from a mercury arc whose wavelengths were independently measured by Walter T. Whitney by grating photograph. Table 1 lists the lines and their frequencies (computed from $\nu = c/\lambda$ with $c = 2.998 \times 10^{10}$ cm/s).

4.2 Millikan's Raw Stopping-Potential Data

Table 2 reproduces the control readings near the balance point (zero electrometer deflection) from Millikan's Table I. The electrometer sensitivity was 2500 mm/V; the Weston voltmeter was certified by the National Bureau of Standards to $\pm 0.1\%$.

The balance points interpolated from these curves (Millikan's Fig. 5) are the stopping potentials used in the regression.

Table 1: Mercury-arc spectral lines used by [Millikan \[1916\]](#). Frequencies computed from $\nu = c/\lambda$, $c = 2.998 \times 10^{10}$ cm/s.

λ (Å)	λ (nm)	ν (10^{14} Hz)
5461.0	546.10	5.490
4339.4	433.94	6.908
4046.8	404.68	7.409
3650.2	365.02	8.213
3125.5	312.55	9.591
2534.7	253.47	11.827

Table 2: Selected readings from Millikan (1916), Table I. Sodium surface. “V” is the applied potentiometer voltage; “mm” is the electrometer deflection in 30 s. For lines 5461–3126 Å the contact EMF ($v_c = 2.51$ V) exceeds the photopotential and the reported voltages are the *assisting* potential near balance (balance = zero deflection). For line 2535 Å voltages are retarding.

5461 Å		4339 Å		4047 Å		3650 Å		3126 Å		2535 Å	
V	mm	V	mm	V	mm	V	mm	V	mm	V	mm
2.257	28	1.581	44	1.576	82	1.157	67.5	0.581	52	−0.058	68
2.205	14	1.629	20	1.524	55	1.105	36	0.529	29	+0.058	38
2.152	7	1.576	10	1.471	36	1.053	19	0.477	12	+0.162	26
2.100	3	1.524	4	1.419	24	1.000	11	0.424	5	+0.267	16.5
		1.367	3	0.948	4			0.372	2.5	+0.372	8

4.3 Reconstructed Stopping Potentials

Table 3 collects the true stopping potentials $V_{0,\text{true}} = V_{0,\text{obs}} + v_c$ for sodium and lithium. For sodium, the contact EMF was independently measured as $v_c = 2.51$ V (stable to ± 0.01 V over 9 hours). For lithium (first series), $v_c = 1.52$ V. For sodium at 2535 Å the paper explicitly states $V_{0,\text{obs}} = 0.52$ V, giving $V_{0,\text{true}} = 3.03$ V (a small scattered-light correction accounts for the 0.04 V discrepancy with the regression-implied value of 3.07 V).

Table 3: Historical stopping-potential data used in the regression. True stopping potentials $V_0 = V_{0,\text{obs}} + v_c$ for sodium (Na, $v_c = 2.51$ V) and lithium (Li, $v_c = 1.52$ V). These values are reconstructed from Millikan (1916) as described in Section 4. The dash for Li at 5461 Å indicates that this wavelength is above lithium’s photoemission threshold ($\lambda_0 \approx 5260$ Å).

λ (Å)	λ (nm)	ν (10^{14} Hz)	V_0^{Na} (V)	V_0^{Li} (V)
5461.0	546.10	5.490	0.454	—
4339.4	433.94	6.908	1.039	0.499
4046.8	404.68	7.409	1.246	0.706
3650.2	365.02	8.213	1.578	1.037
3125.5	312.55	9.591	2.147	1.606
2534.7	253.47	11.827	3.030	2.529

5 OLS Regression: Estimating h/e

5.1 Regression Setup

The model is

$$V_{0i} = \alpha + \beta \nu_i + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} (0, \sigma^2), \quad i = 1, \dots, n.$$

OLS minimises $\sum_{i=1}^n (V_{0i} - \alpha - \beta \nu_i)^2$. The closed-form estimates are

$$\hat{\beta} = \frac{S_{\nu V}}{S_{\nu\nu}} = \frac{\sum_i (\nu_i - \bar{\nu})(V_{0i} - \bar{V}_0)}{\sum_i (\nu_i - \bar{\nu})^2}, \quad (5)$$

$$\hat{\alpha} = \bar{V}_0 - \hat{\beta} \bar{\nu}. \quad (6)$$

5.2 Results: Sodium (Five Points, $\lambda = 5461\text{--}3126$ Å)

Millikan’s primary regression used the five most reliable sodium points (omitting 2535 Å where a scattered-light correction was required). I reproduce this regression exactly.

Data.

ν (10^{14} Hz)	5.490	6.908	7.409	8.213	9.591
V_0 (V)	0.454	1.039	1.246	1.578	2.147

Summary statistics.

$$n = 5,$$

$$\bar{\nu} = \frac{1}{5}(5.490 + 6.908 + 7.409 + 8.213 + 9.591) = 7.522 \times 10^{14} \text{ Hz},$$

$$\bar{V}_0 = \frac{1}{5}(0.454 + 1.039 + 1.246 + 1.578 + 2.147) = 1.293 \text{ V}.$$

Sums of squares and cross-products.

i	$\nu_i - \bar{\nu}$ (10^{14} Hz)	$V_{0i} - \bar{V}_0$ (V)	$(\nu_i - \bar{\nu})^2$	$(\nu_i - \bar{\nu})(V_{0i} - \bar{V}_0)$
1	-2.032	-0.839	4.129	1.705
2	-0.614	-0.254	0.377	0.156
3	-0.113	-0.047	0.013	0.005
4	+0.691	+0.285	0.477	0.197
5	+2.069	+0.854	4.281	1.767
$\sum$			$S_{\nu\nu} = 9.277$	$S_{\nu V} = 3.830$

(Units: $S_{\nu\nu}$ in $(10^{14} \text{ Hz})^2$; $S_{\nu V}$ in $10^{14} \text{ Hz}\cdot\text{V}$.)

OLS estimates.

$$\hat{\beta} = \frac{S_{\nu V}}{S_{\nu\nu}} = \frac{3.830}{9.277} = \mathbf{0.4129} \times 10^{-14} \text{ V/Hz} = \mathbf{4.129} \times \mathbf{10^{-15}} \text{ V}\cdot\text{s}, \quad (7)$$

$$\hat{\alpha} = \bar{V}_0 - \hat{\beta} \bar{\nu} = 1.293 - 4.129 \times 10^{-15} \times 7.522 \times 10^{14} = 1.293 - 3.106 = \mathbf{-1.813} \text{ V}. \quad (8)$$

Residuals and goodness of fit.

λ (Å)	V_{0i} (V)	$\hat{V}_{0i}$ (V)	$\hat{\epsilon}_i$ (V)
5461	0.454	0.454	+0.000
4339	1.039	1.040	-0.001
4047	1.246	1.247	-0.001
3650	1.578	1.578	+0.000
3126	2.147	2.145	+0.002

$R^2 = 1 - \hat{\sigma}^2 n / (n - 1) / S_{VV} \approx 0.99999$. Residuals are below 2 mV, consistent with Millikan's stated uncertainty of ± 0.01 V.

Uncertainty estimate. The residual standard deviation is

$$\hat{\sigma} = \sqrt{\frac{\sum \hat{\varepsilon}_i^2}{n-2}} = \sqrt{\frac{(0^2 + 0.001^2 + 0.001^2 + 0^2 + 0.002^2)}{3}} \approx 0.0014 \text{ V}.$$

The standard error of the slope is

$$\text{SE}(\hat{\beta}) = \frac{\hat{\sigma}}{\sqrt{S_{\nu\nu}}} = \frac{0.0014 \times 10^0}{9.277^{1/2} \times 10^{14}} = \frac{0.0014}{3.046 \times 10^{14}} \approx 4.6 \times 10^{-18} \text{ V}\cdot\text{s},$$

corresponding to a **relative uncertainty of about 0.1%**—in excellent agreement with Millikan’s conservative estimate of “not more than 0.5%” based on graphical fitting.

5.3 Results: Sodium (Six Points, Including 2535 Å)

Including the 2535 Å point ($\nu = 11.827 \times 10^{14} \text{ Hz}$, $V_0 = 3.030 \text{ V}$ —using the explicitly stated true stopping potential) modestly tightens the slope estimate:

$$\begin{aligned} \bar{\nu} &= 8.240 \times 10^{14} \text{ Hz}, & \bar{V}_0 &= 1.582 \text{ V}, \\ S_{\nu\nu} &= 28.14 \times (10^{14})^2, & S_{\nu V} &= 11.64 \times 10^{14} \text{ V}\cdot\text{Hz}, \\ \hat{\beta} &= \frac{11.64}{28.14} = 4.135 \times 10^{-15} \text{ V}\cdot\text{s}, \end{aligned}$$

a change of less than 0.2% from the five-point estimate.

5.4 Results: Lithium (Five Points)

Repeating for lithium using lines 4339–2535 Å:

$$\hat{\beta}_{\text{Li}} = 4.137 \times 10^{-15} \text{ V}\cdot\text{s},$$

consistent with the sodium result to four significant figures, confirming the universality of the slope.

5.5 Deriving Planck’s Constant

With the OLS slope $\hat{\beta} = 4.129 \times 10^{-15} \text{ V}\cdot\text{s}$ (five-point sodium regression) and the elementary charge e from Millikan’s oil-drop experiment:

Value of e used	Implied $h = \hat{\beta} \cdot e$	Error
Millikan (1916): $1.592 \times 10^{-19} \text{ C}$	$6.573 \times 10^{-34} \text{ J}\cdot\text{s}$	-0.80%
Modern NIST 2022: $1.6022 \times 10^{-19} \text{ C}$	$6.614 \times 10^{-34} \text{ J}\cdot\text{s}$	$+0.49\%$
Modern exact: $h = 6.6261 \times 10^{-34} \text{ J}\cdot\text{s}$	—	0

The small positive error when using the modern e reflects rounding in the tabulated stopping potentials. Millikan’s negative error arose primarily from his underestimate of e (by 0.65%).

6 Maximum Likelihood Estimation of Einstein’s Parameters

6.1 The Parametric Model

Einstein’s photoelectric equation defines a three-parameter statistical model. The free parameters are:

- $\beta = h/e$ — the universal slope ($\text{V}\cdot\text{s}$), encoding Planck’s constant;
- $\alpha = v_c - W/e$ — a metal- and run-specific intercept (V), conflating the work function and contact EMF;
- σ^2 — the variance of the measurement noise ε_i .

The full parametric model for a single metal over n spectral lines is:

$$V_{0i} = \alpha + \beta \nu_i + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2), \quad i = 1, \dots, n. \quad (9)$$

The Gaussian assumption is natural: measurement noise in the electrometer readings arises from many small independent sources (thermal Johnson noise, vibration, residual-

gas collisions), for which a central-limit argument is compelling.

6.2 The Log-Likelihood

The probability density of a single observation (V_{0i}, ν_i) under model (9) is

$$p(V_{0i} \mid \nu_i; \alpha, \beta, \sigma) = \frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{(V_{0i} - \alpha - \beta\nu_i)^2}{2\sigma^2}\right).$$

For n independent observations the log-likelihood is

$$\ell(\alpha, \beta, \sigma) = -\frac{n}{2} \ln(2\pi) - n \ln \sigma - \frac{1}{2\sigma^2} \underbrace{\sum_{i=1}^n (V_{0i} - \alpha - \beta\nu_i)^2}_{\text{RSS}(\alpha, \beta)}. \quad (10)$$

6.3 MLE Equals OLS Under Gaussian Errors

For fixed σ , maximising (10) over (α, β) requires minimising $\text{RSS}(\alpha, \beta)$ — exactly the OLS objective. Therefore:

$$\boxed{(\hat{\alpha}_{\text{MLE}}, \hat{\beta}_{\text{MLE}}) = (\hat{\alpha}_{\text{OLS}}, \hat{\beta}_{\text{OLS}}).} \quad (11)$$

The equivalence holds for any σ .

6.4 MLE for the Noise Variance

Substituting the maximising $(\hat{\alpha}, \hat{\beta})$ into (10) and optimising over σ gives:

$$\hat{\sigma}_{\text{MLE}}^2 = \frac{\text{RSS}}{n}. \quad (12)$$

The OLS estimator uses the degrees-of-freedom correction:

$$s^2 = \frac{\text{RSS}}{n-2}, \quad (13)$$

which is unbiased ($\mathbb{E}[s^2] = \sigma^2$). The MLE $\hat{\sigma}_{\text{MLE}}^2$ is biased downward by the factor $(n-2)/n$; for $n=5$ the correction is $3/5$.

For the five-point sodium regression, $\text{RSS} = 6 \times 10^{-6} \text{ V}^2$ (from the residuals in Section 5).

Hence:

$$\begin{aligned}\hat{\sigma}_{\text{MLE}} &= \sqrt{6 \times 10^{-6}/5} = 1.1 \times 10^{-3} \text{ V} = 1.1 \text{ mV}, \\ s_{\text{OLS}} &= \sqrt{6 \times 10^{-6}/3} = 1.4 \times 10^{-3} \text{ V} = 1.4 \text{ mV}.\end{aligned}$$

Both estimates lie far below Millikan's stated experimental uncertainty of $\pm 10 \text{ mV}$, confirming that the residuals are dominated by rounding in the tabulated stopping potentials rather than by true measurement scatter.

6.5 Fisher Information and the Cramér-Rao Bound

The Fisher information matrix bounds the precision attainable by any unbiased estimator of (α, β) . For model (9) with σ known, the Fisher information is

$$\mathcal{I}(\alpha, \beta) = \frac{1}{\sigma^2} X^\top X = \frac{1}{\sigma^2} \begin{pmatrix} n & \sum_i \nu_i \\ \sum_i \nu_i & \sum_i \nu_i^2 \end{pmatrix}, \quad (14)$$

where the row $(1, \nu_i)$ of the $n \times 2$ design matrix X maps (α, β) into the conditional mean of V_{0i} . The Cramér-Rao bound on the variance of any unbiased estimator of β is

$$\text{Var}(\tilde{\beta}) \geq [\mathcal{I}^{-1}(\alpha, \beta)]_{\beta\beta} = \frac{\sigma^2}{S_{\nu\nu}}, \quad (15)$$

where $S_{\nu\nu} = \sum_i (\nu_i - \bar{\nu})^2$. The OLS/MLE estimator $\hat{\beta}$ achieves this bound: $\text{Var}(\hat{\beta}) = \sigma^2/S_{\nu\nu}$ (by the Gauss-Markov theorem, OLS is the Best Linear Unbiased Estimator).

For the five-point sodium regression (using s as a plug-in for σ):

$$\text{SE}_{\text{CR}}(\hat{\beta}) = \frac{s}{\sqrt{S_{\nu\nu}}} = \frac{1.4 \times 10^{-3} \text{ V}}{\sqrt{9.277 \times 10^{14} \text{ Hz}}} = 4.6 \times 10^{-18} \text{ V}\cdot\text{s}, \quad (16)$$

corresponding to a relative precision of 0.11%, which OLS achieves exactly.

Design implications. By (15), $\text{SE}(\hat{\beta}) \propto 1/\sqrt{S_{\nu\nu}}$. Spreading the frequencies further apart shrinks the standard error whatever the value of n . This justifies Millikan's choice to span nearly a factor of two in ν (5461 Å to 2535 Å), rather than the narrow 60 nm UV ranges used by Hughes and Richardson & Compton.

Table 4 quantifies the improvement. Restricting Millikan’s data to the 4047–2535 Å range (as in Hughes) gives $S_{\nu\nu} = 1.96 \times (10^{14})^2 \text{ Hz}^2$ vs. the full-range $9.277 \times (10^{14})^2 \text{ Hz}^2$ — a factor of 4.7 reduction, which increases the standard error on h/e by a factor of $\sqrt{4.7} \approx 2.2$.

Table 4: Effect of frequency range on the Cramér-Rao bound for $\hat{\beta} = h/e$, using Millikan’s sodium data. “Restricted” uses only the three lines nearest to the Hughes (1912) wavelength range.

Dataset	n	$S_{\nu\nu} (10^{28} \text{ Hz}^2)$	$S_{\nu\nu}^{-1/2} (\text{rel.})$	$\text{SE}(\hat{\beta}) (10^{-18} \text{ V}\cdot\text{s})$
Restricted (4047–2535 Å)	3	1.96	1.00	10.0
Millikan 5-pt (5461–3126 Å)	5	9.277	0.46	4.6
Millikan full (5461–2535 Å)	6	28.14	0.26	2.6

6.6 Pooled Estimation Across Metals

A more efficient estimator exploits the cross-metal restriction: $\beta = h/e$ is universal. The pooled model is

$$V_{0im} = \alpha_m + \beta \nu_i + \varepsilon_{im}, \quad m \in \{\text{Na}, \text{Li}\}, \quad (17)$$

where the intercept α_m is specific to metal m and the slope β is common to Na and Li. The pooled OLS slope estimator is

$$\hat{\beta}_{\text{pool}} = \frac{S_{\nu V}^{\text{Na}} + S_{\nu V}^{\text{Li}}}{S_{\nu\nu}^{\text{Na}} + S_{\nu\nu}^{\text{Li}}}. \quad (18)$$

From Section 5, $\hat{\beta}_{\text{Na}} = 4.129$ and $\hat{\beta}_{\text{Li}} = 4.137$ (both $\times 10^{-15} \text{ V}\cdot\text{s}$), differing by 0.19%. A pooled estimate weighting by $S_{\nu\nu,m}$ gives $\hat{\beta}_{\text{pool}} \approx 4.133 \times 10^{-15} \text{ V}\cdot\text{s}$, with a standard error reduced by roughly $1/\sqrt{2}$ relative to either single-metal estimate.

The Wald statistic for testing the slope-equality restriction $H_0: \beta_{\text{Na}} = \beta_{\text{Li}}$ is

$$z = \frac{\hat{\beta}_{\text{Na}} - \hat{\beta}_{\text{Li}}}{\sqrt{\text{SE}^2(\hat{\beta}_{\text{Na}}) + \text{SE}^2(\hat{\beta}_{\text{Li}})}} \approx \frac{0.008 \times 10^{-15}}{2 \times 4.6 \times 10^{-18}} \approx 0.87.$$

The test statistic is well below the critical value of 1.96, so the data provide no evidence against universality of h/e across sodium and lithium — a confirmation of a key

structural prediction of the quantum hypothesis.

7 The Contact-EMF Identification Problem

The contact EMF v_c between the photoemitting metal and the collector is a nuisance parameter. The intercept $\alpha = -W/e + v_c$ of (4) conflates the work function W/e with the apparatus term v_c .

Pre-Millikan experiments. The early experiments did not measure or control v_c . If v_c drifted between measurements at different frequencies, the apparent stopping potentials would shift inconsistently and bias both slope and intercept. Hughes and Richardson & Compton measured different lines on different days with no guarantee of constant v_c , which Millikan [1916] identified as a contributing cause of their h values being 10–45% below Planck’s.

Millikan’s solution. Millikan measured all six spectral lines in a single continuous session after a freshly shaved surface had stabilised (typically 1–2 hours after shaving). He verified that v_c remained constant to ± 0.01 V over the session by using the Kelvin double-balance. When v_c is constant across all observations, it enters as a pure intercept shift and does not affect $\hat{\beta}$. The slope $\hat{\beta} = h/e$ is then estimated without bias regardless of the unknown value of v_c .

Separate identification of W . By independently measuring $v_c = 2.51$ V (Kelvin method) and reading the intercept $\hat{\alpha} = -1.81$ V from the regression (equation (8)), Millikan recovered

$$\begin{aligned} W_{\text{Na}}/e &= v_c - \hat{\alpha} = 2.51 - (-1.81) = 4.32 \text{ eV}, \\ W_{\text{Na}} &= 4.32 \times 1.602 \times 10^{-19} = 6.92 \times 10^{-19} \text{ J}. \end{aligned}$$

However, he noted that the exact work function of a freshly scraped sodium surface in 10^{-2} mmHg residual gas was not well characterised. The widely quoted value for sodium is $W_{\text{Na}} = 2.36$ eV (modern value for clean bulk Na). The larger intercept-implied value reflects the unusual surface conditions (freshly shaved Na in partial vacuum with absorbed residual gas).

8 Nine Two-Point Determinations

In addition to the direct regression, Millikan independently estimated the slope from nine pairs of spectral lines, always pairing one line with 5461 Å (chosen for its high intensity and hence most precise balance point).

Table 5: Millikan (1916), Table II: nine two-point slope determinations for sodium. Each row is an independent measurement session with a fresh surface shaving. The mean and standard deviation are computed from these nine observations.

Session	λ paired with 5461 Å	$\hat{\beta}$ (10^{-15} V·s)
1	3126 Å	4.11
2	3650 Å	4.14
3	3126 Å	4.10
4	3650 Å	4.12
5	3126 Å	4.24
6	4047 Å	3.98
7	2535 Å	4.04
8	3126 Å	4.24
9	4047 Å	4.21
Mean		4.131
Std. dev. ($n = 9$)		0.087
Std. error of mean		0.029
95% CI		[4.06, 4.20]

The two-point mean 4.131×10^{-15} V·s agrees with the five-point OLS estimate 4.129×10^{-15} V·s to better than 0.1%. The variation across sessions (sessions 5 and 8 giving 4.24 vs. sessions 3 and 6 giving 3.98–4.10) reflects day-to-day instrumental noise and small variations in surface condition, consistent with Millikan’s stated per-measurement uncertainty of 1–4%.

9 Comparison of Historical Estimates

Table 6 compares the estimates of h from all major pre-1920 photoelectric experiments.

All pre-Millikan estimates are biased downward by 10–47%. The bias is attributable to stray short-wavelength light (raises apparent V_0 at long-wavelength lines, flattening the slope), restricted frequency range (enlarges SE of slope), and uncontrolled contact EMF

Table 6: Historical estimates of Planck’s constant h from photoelectric experiments. Millikan’s attribution of earlier errors is discussed in [Millikan \[1916, pp. 373–375\]](#). “Error” is relative to the modern NIST CODATA 2022 value $h = 6.6261 \times 10^{-27}$ erg·s.

Authors	Year	Metal(s)	h (10^{-27} erg·s)	Error
Hughes	1912–13	various	3.55–5.85	–12 to –46%
Richardson & Compton	1912	8 metals	3.5–5.8	–12 to –47%
Millikan (Na, 5-pt)	1916	Na	6.573	–0.80%
Millikan (Na, 9-pt)	1916	Na	6.578	–0.73%
Millikan (Li)	1916	Li	6.584	–0.64%
Millikan (Na+Li mean)	1916	Na+Li	6.579	–0.72%
Planck (black body)	1913	—	6.415	–3.19%
Modern NIST 2022	2022	—	6.6261	0

(adds noise and possible trend bias to the intercept). Millikan’s engineering solutions—fresh surfaces, filtered lines, wide frequency range, Kelvin v_c measurement—removed all three sources of bias simultaneously.

10 Millikan’s Reluctance

Millikan’s 1916 paper confirmed Einstein’s equation (1) to within 0.5%. Yet Millikan wrote:

“Despite then the apparently complete success of the Einstein equation, the physical theory of which it was designed to be the symbolic expression is found so untenable that Einstein himself, I believe, no longer holds to it.”
[\[Millikan, 1916, p. 388\]](#)

The light-quantum hypothesis was, in Millikan’s judgment, physically implausible because it conflicted with the wave theory of diffraction and interference. He had hoped his experiment would *refute* the equation, and was surprised when it confirmed it.

By 1923—after Compton scattering [\[Compton, 1923\]](#) provided independent evidence for the particulate nature of light—Millikan conceded:

“This work resulted, contrary to my own expectation, in the first direct experimental proof . . . of the complete validity of the Einstein equation.”
(Nobel address, 1923; quoted by [Holton 1999](#))

A macroeconomist likewise can confirm a reduced-form predictive relationship without endorsing the structural model that motivated it.

11 Einstein as Machine Learner

11.1 Overview

The vocabulary of modern machine learning—supervised learning, hypothesis class, loss function, inductive bias, feature engineering, transfer learning—did not exist in 1905. The operations Einstein performed are the ones that statistical learning theory formalises. This section states the correspondence and its limits.

11.2 Supervised Learning and Labeled Data

The photoelectric dataset is a standard supervised regression problem. The *input feature* is the frequency ν of the incident light; the *label* is the stopping potential V_0 . The task is to learn a function $f : \nu \mapsto V_0$ that generalises to unobserved frequencies.

Lenard (1902) provided the *qualitative* version: the label was observed (high V_0 at high frequency) but not quantified reliably. Millikan (1916) provided the *quantitative* training set: six labeled pairs (ν_i, V_{0i}) for each metal, with measurement precision sufficient to estimate the slope h/e to 0.5%.

11.3 The Hypothesis Class and Inductive Bias

Einstein’s hypothesis class is:

$$\mathcal{H} = \{f_{\alpha,\beta} : \nu \mapsto \alpha + \beta\nu \mid \alpha \in \mathbb{R}, \beta > 0\}, \quad (19)$$

together with the *cross-metal constraint* that β is the same for all metals. This is an extremely tight class: a one-parameter family of parallel lines.

The restriction to $\mathcal{H}$ was not ad hoc. It followed deductively from the quantum hypothesis $E = h\nu$ and energy conservation for photoemission. The quantum hypothesis *is* the inductive bias: it restricts the hypothesis class from all possible functions of $(\nu, I, \lambda, \dots)$ to a specific affine form in ν alone. Without this bias five data points are hopelessly insufficient: infinitely many smooth functions pass through five points.

11.4 Feature Engineering: Choosing ν Over Intensity and Wavelength

Learning begins with a choice of input features. Lenard’s experiments performed a feature importance analysis. He measured V_0 as a function of both frequency (colour) and intensity I . His result was unambiguous: V_0 varies strongly with ν but not at all with I .

Classical wave theory predicted the opposite: wave amplitude $A \propto \sqrt{I}$ should determine energy delivered to an electron, so V_0 should grow with I . The empirical irrelevance of I was the anomaly that motivated Einstein’s model. Lenard’s experiment eliminated I from the feature set and selected ν before any parametric fitting was done.

11.5 The Loss Function and Training

Given the hypothesis class (19) and the feature ν , choose (α, β) to minimise the empirical risk under squared-error loss:

$$\hat{R}(\alpha, \beta) = \frac{1}{n} \sum_{i=1}^n (V_{0i} - \alpha - \beta \nu_i)^2.$$

This is identical to the MSE loss used in neural-network regression. The minimiser of $\hat{R}$ is the OLS/MLE estimator derived in Sections 5–6: training is the closed-form solution $\hat{\beta} = S_{\nu V} / S_{\nu \nu}$. For Einstein’s linear model, the loss surface is a convex paraboloid in (α, β) with a unique global minimum; no stochastic gradient descent is needed.

11.6 Generalisation and Transfer Learning

Generalisation. The fitted line $\hat{V}_0 = \hat{\alpha} + \hat{\beta} \nu$ predicts stopping potentials for frequencies *not in the training set*. The near-zero residuals (Section 5) show no overfitting. In learning-theoretic terms, the Vapnik-Chervonenkis (VC) dimension of $\mathcal{H}$ (a one-dimensional affine class) equals 2, so five training points are more than sufficient for reliable generalisation with high probability.

Transfer learning. A model trained on sodium data successfully predicts lithium stopping potentials up to an intercept shift. The slope $\hat{\beta} = h/e$ is transferred exactly from one metal to another. Training on sodium provides a pre-trained feature extractor,

the slope; the lithium task fine-tunes the intercept α_{Li} , the metal-specific work-function parameter. The Wald test of Section 6.6 confirms that the zero-shot transfer of the slope from Na to Li fails to be rejected by the data ($z = 0.87$).

11.7 Model Selection: Quantum vs. Classical Wave Theory

At the time of Einstein’s paper two models were in competition:

- **Classical wave model.** Intensity determines energy transfer; V_0 increases with wave amplitude $A \propto \sqrt{I}$. This predicts zero ν -dependence in V_0 , i.e., $R^2 = 0$ on the (ν, V_0) data.
- **Quantum model.** $V_0 = (h/e)\nu - W/e$. This predicts an exact linear ν -dependence with a specific slope that is universal across metals.

Millikan’s data delivers $R^2 \approx 0.99999$ for the quantum model. Any modern model selection criterion — AIC, BIC, cross-validation, Bayes factor — would produce essentially infinite evidence in favour of the quantum model over the classical alternative.

11.8 What a Naive Machine Learning System Would Miss

A purely data-driven system, given only the table of (ν_i, V_{0i}) pairs, would learn the following.

1. A linear regression package would correctly recover $\hat{\beta} = 4.129 \times 10^{-15} \text{ V}\cdot\text{s}$ and $R^2 \approx 1$.
2. It would *not* know that $\hat{\beta}$ identifies a fundamental constant of nature. The number 4.129×10^{-15} is physically meaningless without the quantum hypothesis that gives it the interpretation h/e .
3. It would *not* impose the cross-metal universality restriction $\beta_{\text{Na}} = \beta_{\text{Li}}$ from first principles. The universality is a prediction of the quantum hypothesis; from a purely data-driven perspective, it is just a coincidence to be discovered, not a constraint to be imposed.
4. Modern symbolic regression systems such as AI Feynman [Udrescu and Tegmark, 2020] can, in principle, rediscover the linear form from data. But they require knowledge of units and physical constants to identify $\hat{\beta}$ with h/e rather than

reporting it as an unexplained numerical coefficient.

Einstein supplied the theoretical framework that gives the regression coefficients their physical meaning. Specifying the hypothesis class required more intellectual effort than fitting the line.

11.9 The Deductive–Inductive Loop

Modern discussions of “AI for science” often describe a purely inductive paradigm: given enough data, learn the law. The photoelectric episode follows the hypothetico-deductive loop that has driven physical science since Newton.

Table 7: The hypothetico-deductive loop in the photoelectric episode, translated into the language of modern machine learning.

Step	Scientific name	ML name
Propose $E = h\nu$	Theoretical hypothesis	Model architecture
Derive $V_0 = (h/e)\nu - W/e$	Deduction	Hypothesis class
Choose ν not I	Feature selection	Feature engineering
Measure (ν_i, V_{0i})	Experiment	Training data
Min. $\sum (V_{0i} - \hat{V}_{0i})^2$	Least squares	MSE training
$ \hat{\beta} - h/e < 1\%$	Confirmation	Test accuracy
Na and Li: same $\hat{\beta}$	Universality	Zero-shot transfer

Five training points estimate h to 0.1% because the physics prescribes the functional form, the number of free parameters (α, β) , and the units of the slope. A purely inductive system that must discover the functional form from data faces a far larger sample-complexity problem.

12 A Deep Neural Network on the Photoelectric Data

12.1 Two Hypothesis Classes Compared

Einstein’s model and a deep neural network (DNN) represent opposite extremes of the bias-variance spectrum for the photoelectric regression. Einstein’s hypothesis class is equation (19): a one-parameter family of parallel lines with VC dimension equal to 2.

The Gauss–Markov theorem and the Cramér–Rao bound together guarantee that OLS achieves minimum variance among all unbiased estimators; five training points are more than sufficient.

A DNN with L hidden layers and N neurons per layer is a universal function approximator: for any continuous target function on a compact interval, there exists a network that approximates it to arbitrary precision [Hornik et al., 1989]. For a one-dimensional input (ν) and one-dimensional output (V_0), even a modest architecture—three hidden layers with 16 ReLU neurons each—has approximately

$$p = (1 \cdot 16 + 16) + (16 \cdot 16 + 16) + (16 \cdot 16 + 16) + (16 \cdot 1 + 1) = 593$$

free parameters. The VC dimension of this network is of order $p \log p \approx 5,700$ [Vapnik, 1998]. Reliable generalisation to unseen frequencies then requires $n \gtrsim 5,700$ labeled training examples for 1% accuracy with high probability. The five points Millikan assembled in 1916 fall short by three orders of magnitude.

12.2 Epoch I—Lenard’s Qualitative Findings (1902)

Lenard established two qualitative facts: stopping potential rises with frequency and is independent of intensity. There are no numerical (ν_i, V_{0i}) pairs. A DNN requires labeled numerical examples; with none, it cannot begin training. Einstein required zero labeled examples: the quantum hypothesis plus energy conservation yielded equation (1) as a deduction. The asymmetry is stark—Einstein’s “model fitting” was a one-paragraph derivation; the DNN’s fitting cannot start.

12.3 Epoch II—Hughes and Richardson & Compton (1907–1914)

The intermediate experiments produced roughly 25–40 numerical (ν_i, V_{0i}) pairs, clustered in a narrow 60 nm UV range with uncontrolled contact EMFs and stray scattered light. Three problems foreclose reliable DNN learning.

Underdetermination. With $n \approx 30$ and $p = 593$, the DNN is massively overparameterised. Gradient descent finds a minimum-norm interpolant that passes through all 30 training points exactly (zero training loss), but is effectively unconstrained between

them. Applying ℓ_2 weight decay suppresses oscillations, but the optimal regularisation strength cannot be selected without a held-out validation set; with $n = 30$ this is statistically isolated.

Restricted frequency range. A linear, a quadratic, and a square-root form fit equally well over a 60 nm UV band ($\Delta\nu \approx 0.7 \times 10^{14}$ Hz). The DNN has no mechanism to infer which of them is the global functional form.

Systematic label contamination. Uncontrolled contact EMFs imply that the effective label for observation i is $V_{0i} + \delta_i$, where δ_i is unknown and observation-specific. The DNN receives contaminated targets and, lacking any physical model of the contaminant, cannot separate the linear physical trend from the systematic noise. The learned function absorbs both, producing a biased and uninterpretable output.

12.4 Epoch III—Millikan’s Data (1916)

Millikan’s five-point sodium dataset is the cleanest photoelectric dataset of the era.

The interpolation regime. With $n = 5$ and any DNN with $p > 5$ (which includes every useful architecture), the network operates in the interpolation regime: it memorises all five training points exactly, producing zero training loss [Belkin et al., 2019]. The resulting function is unconstrained everywhere between the training points. Even the best-case minimum-norm interpolant—“well-behaved” in the double-descent sense—provides no basis for slope inference: there are no residual degrees of freedom with which to estimate the noise variance or a standard error on the slope.

Loss of inferential machinery. OLS with $n = 5$ and $p = 2$ free parameters uses $n - 2 = 3$ residual degrees of freedom to estimate $\hat{\sigma}^2$ and hence $\text{SE}(\hat{\beta})$. Any DNN with more than three parameters exhausts those degrees of freedom entirely. The DNN cannot report a confidence interval for h/e , cannot test whether the slope differs from zero, and cannot quantify the 0.1% relative precision that OLS achieves at the Cramér–Rao bound.

No basis for universality. The universality of the slope $\beta = h/e$ across metals is a theoretical prediction of the quantum hypothesis, not an empirical feature discoverable

from sodium data alone. A DNN trained on sodium would need to be retrained from scratch on lithium. The only mechanism by which the slope could transfer automatically from Na to Li is an explicit architectural constraint that encodes the universality—which is equivalent to importing the physics into the model.

12.5 The Modern Era: Large Datasets

By the mid-twentieth century, photoelectric measurements spanned decades of metals, surface preparations, and photon energies. Pooling even a modest subset—say $n = 5,000$ clean observations over two octaves of frequency for a single metal—places a DNN firmly in the classical regime ($n \gg p$).

With smooth activations and appropriate regularisation, the DNN would converge, as $n \rightarrow \infty$, to the minimum-population-risk function in its hypothesis class. Because the true function is exactly linear, the minimum-risk function in any sufficiently rich class *is* the linear function. The DNN would eventually recover the line. What it still could not do:

- Identify the learned slope $\hat{w} \approx 4.129 \times 10^{-15} \text{ V}\cdot\text{s}$ as the ratio h/e of two independently measured constants. The slope appears as a numerical weight, not as a ratio of fundamental quantities.
- Impose the cross-metal constraint $\beta_{\text{Na}} = \beta_{\text{Li}} = \dots$ from first principles. Universality would appear as an empirical pattern to be confirmed, not a structural prior to be imposed.
- Decompose the intercept into a physical work function W/e and an apparatus nuisance v_c without independent measurements of v_c .
- Reduce the required sample from $O(10^3)$ examples to five. A correctly specified two-parameter linear model needs five points; no architectural innovation can replicate this compression without importing the physics.

Modern symbolic regression systems such as AI Feynman [Udrescu and Tegmark, 2020] can, in principle, rediscover the linear form and even match $\hat{w}$ to h/e if given units and physical constants as hardcoded inputs. But that hardcoding is itself a form of prior physical knowledge—the quantum hypothesis in disguise—not something extracted from the (ν, V_0) data.

12.6 Summary: Sample Complexity and the Value of Theory

Table 8 quantifies the contrast.

Table 8: Sample-complexity comparison between Einstein’s two-parameter linear model (VC dimension = 2, derived from the quantum hypothesis) and a three-layer DNN with 16 neurons per layer ($p = 593$ parameters, VC dim $\approx 5,700$). The “sufficient n ” for a 1% generalisation error at 95% confidence is $O(\text{VC dim}/\epsilon^2)$.

Epoch	n available	Einstein model	DNN (VC $\approx$ 5700)	DNN outcome
Lenard 1902	≈ 0	Sufficient [†]	Insufficient	Cannot train (no labels)
Hughes 1912	~ 30	Sufficient	Insufficient ($\times 190$)	Memorises; no generalisation
Millikan 1916	5	Sufficient	Catastrophically insufficient	Interpolates; no inference
Modern era	~ 5000	Sufficient	Sufficient	Recovers linear shape; cannot identify h/e

[†] The functional form is derived from theory; only two free parameters remain to be estimated from data.

DNNs are good learners for high-dimensional problems in which no parsimonious parametric family is known: image recognition, protein structure prediction, language modelling. The photoelectric problem sits at the opposite extreme: a one-dimensional input, a one-dimensional output, and five to thirty training examples.

Einstein’s proposal $E = h\nu$ reduced the hypothesis space from the infinite-dimensional function class of a DNN to a two-dimensional manifold of parallel lines, and with it the required number of training examples by three orders of magnitude. He did so before a single labeled example was available.

13 Conclusions

Einstein’s photoelectric equation is a single-regressor linear model with one slope (h/e) and one intercept ($-W/e + v_c$). The central estimation challenge is not the regression itself—which is trivial given clean data—but obtaining clean data: fresh metal surfaces free of oxide (eliminates work-function noise), high vacuum (eliminates atmospheric contamination), spectrally pure light (eliminates bias from stray UV), a

wide frequency range (reduces SE of slope), and control of the nuisance parameter v_c (enables identification of both slope and intercept).

Millikan’s decade-long experimental program was a campaign to raise data quality until the OLS standard errors were small enough to distinguish a specific value of the slope ($h/e = 4.136 \times 10^{-15} \text{ V}\cdot\text{s}$) from the alternatives implied by earlier, discrepant theories.

The final five-point sodium regression gives

$$\begin{aligned}\hat{\beta} &= 4.129 \times 10^{-15} \text{ V}\cdot\text{s} \quad (h = 6.61 \times 10^{-34} \text{ J}\cdot\text{s}, \text{ 0.5\% above modern}), \\ \hat{\alpha} &= -1.813 \text{ V} \quad (W_{\text{Na}}/e \approx 4.3 \text{ eV given } v_c = 2.51 \text{ V}), \\ R^2 &\approx 0.99999.\end{aligned}$$

A Einstein’s Equation in Modern Physics

A.1 The Equation Remains Exact

Einstein’s photoelectric equation (1) is not an approximation to a deeper expression—it is an exact consequence of quantum electrodynamics (QED). In QED the photoelectric effect is described as the absorption of a single photon by an electron near the Fermi level of a metal; energy conservation yields equation (1) directly. The work function W is the electrochemical potential required to remove an electron from the Fermi level to the vacuum level; modern density-functional theory computes W to within a few percent for clean surfaces. Millikan’s regression therefore tests an exact result of QED, not an empirical approximation.

The equation is exploited continuously in surface science. Ultraviolet Photoelectron Spectroscopy (UPS) and Kelvin probe instruments apply equation (1) to measure work functions of semiconductor films, organic photovoltaic layers, and heterogeneous catalysts with sub-meV resolution.

A.2 X-ray Photoelectron Spectroscopy

When the incident photon energy $h\nu$ is drawn from the X-ray range (100 eV–10 keV), the emitted electron’s kinetic energy eV_0 identifies the *binding energy* E_B of the core

electron level from which it was ejected [Siegbahn et al., 1967]:

$$E_B = h\nu - eV_0 - \phi_{\text{spec}}, \quad (20)$$

where ϕ_{spec} is the spectrometer work function, a calibration constant. Equation (20) is equation (1) rearranged: the unknown is now the binding energy rather than h/e . Because each element has a unique set of core-level binding energies, X-ray Photoelectron Spectroscopy (XPS/ESCA) provides quantitative, element-specific chemical analysis of surfaces. Kai Siegbahn received the 1981 Nobel Prize in Physics for this technique, which is now one of the most widely used methods in materials science, semiconductor device fabrication, and surface catalysis.

In statistical terms, XPS is a multivariate extension of Millikan’s regression: dozens of binding energies are estimated simultaneously by inverting Einstein’s linear equation at multiple photon energies.

A.3 Multi-Photon Photoelectric Effect

Einstein’s equation assumes each emitted electron absorbs exactly one photon. At the peak intensities of pulsed lasers ($I \gtrsim 10^{10} \text{ W/cm}^2$), an electron can sequentially absorb $n > 1$ photons before escaping the surface. The n -photon generalisation of equation (1) is

$$eV_0 = n h\nu - W, \quad n = 1, 2, 3, \dots \quad (21)$$

Above-threshold ionisation (ATI)—photoelectron emission at kinetic energies exceeding the single-photon threshold by integer multiples of $h\nu$ —was first observed by Agostini et al. [1979]. The resulting photoelectron spectrum displays a comb of peaks separated by $h\nu$, each satisfying equation (21) for the corresponding n . ATI extends hypothesis class (19) to allow non-unit photon multiplicity; the slope remains h/e and the universal structure is preserved across metals.

A.4 Does the Photoelectric Effect Prove the Quantisation of Light?

Lamb and Scully [1969] showed that the photoelectric stopping-potential equation can be derived from a *semiclassical model*: the electromagnetic field is treated as a classical wave while only the atom (electron) is quantised. In this model a classical wave drives

transitions between quantised atomic energy levels, and the ejected-electron kinetic energy still satisfies equation (1). The reason is that equation (1) follows from energy conservation and the quantised *atomic* level structure alone; it does not require the radiation field itself to be quantised.

Millikan’s regression confirms the functional form $V_0 \propto \nu$ and measures h/e to 0.5%. It does not distinguish the quantised-field hypothesis from semiclassical alternatives. The slope h/e gives Planck’s constant whether or not the field is composed of quanta.

A.5 Photon Antibunching: Definitive Evidence for Field Quantisation

The genuine experimental proof that the radiation field must be quantised came not from the photoelectric effect but from photon-counting experiments in quantum optics. The relevant observable is the statistical regularity—or irregularity—with which successive photons are emitted.

[Hanbury Brown and Twiss \[1956\]](#) observed *photon bunching* in thermal (chaotic) light: photon pairs from a thermal source arrive at two detectors with a coincidence rate exceeding a Poisson benchmark, reflecting the Bose-Einstein statistics of thermal photons. [Glauber \[1963\]](#) developed the quantum theory of optical coherence, introducing coherent states of the radiation field (the states most closely resembling classical waves) and deriving the photon-number statistics that separate classical, coherent, and thermal light. Glauber received the 2005 Nobel Prize in Physics for this framework.

The decisive evidence for field quantisation is photon *antibunching*: the observation that successive photons from a single quantum emitter are *less* likely to arrive together than Poisson statistics predict. [Kimble et al. \[1977\]](#) first observed sub-Poissonian photon-number statistics in the resonance fluorescence of individual sodium atoms. Antibunching is impossible in any classical or semiclassical model of the electromagnetic field: a classical field, however modulated or noisy, always produces at minimum Poissonian coincidences. The antibunching experiment rules out the semiclassical hypothesis of [Lamb and Scully](#) and establishes that the radiation field is quantised.

Einstein’s 1905 paper introduced the photon as a *heuristic*—his own word in the title—and Millikan confirmed its equation to 0.5% in 1916. Definitive experimental evidence that the radiation field is itself quantised arrived only in 1977, by a route—

photon-number statistics in resonance fluorescence—that Einstein’s equation neither anticipated nor directly implied. The regression confirmed the equation; quantum optics experiments seventy years later confirmed the physical interpretation.

A.6 Implications for the Machine Learning Reinterpretation

The modern results above bear on Section 11.

Depth of the hypothesis class. Einstein chose hypothesis class (19) by a heuristic argument whose full justification—QED, Fermi’s golden rule, the Fermi-Dirac distribution of electrons in a metal—emerged over the following three decades. The class proved correct for reasons deeper than those available in 1905, and it extends over a five-order-of-magnitude range of photon energies, from visible UV to hard X-rays, without modification.

What the regression tests. Millikan’s regression tests the functional form $V_0 \propto \nu$ and estimates the slope h/e . Both are consistent with the quantised-field and with the semiclassical model of radiation. The regression identifies the best-fitting model within a hypothesis class; selecting between the two classes requires a different observation, photon-number variance, and a different experimental design.

Theory specificity in experimental design. The antibunching experiments of Kimble et al. [1977] were designed specifically to test the quantised-field hypothesis; the relevant observable (photon-number variance) follows directly from Glauber’s coherence theory. Without the quantised-field hypothesis and its prediction of sub-Poissonian statistics, there was no reason to measure photon-number correlations at all. Theory specifies the hypothesis class and, from it, the decisive test. Each theoretical refinement between Millikan (1916) and Kimble et al. (1977) generated a new and more discriminating experiment, a chain of hypothetico-deductive loops (Table 7) whose links are forged by theory, not by data alone.

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